



Research Article

Impact of connectivity characteristics on the permeability of three-dimensional fracture networks

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ABSTRACT

A program for rapid modelling, connectivity evaluation, and seepage simulation of a three-dimensional fracture network was developed based on COMSOL using MATLAB to accurately evaluate the seepage characteristics of a rock fracture network. A Monte Carlo algorithm was used to generate the fracture geometry parameters, and the Euler angle was introduced to establish a three-dimensional fracture model. The bounding box method was used to calculate the fracture network connectivity, and the Darcy module was used to simulate the seepage process. The fracture density, dip direction, dip angle, and length were selected as the experimental factors. Orthogonal numerical experiments were conducted to explore the primary factors controlling the connectivity, permeability, and permeability anisotropy of a three-dimensional fracture network. The results show that connectivity and permeability were primarily affected by the fracture density, length, and dip direction. Moreover, the permeability anisotropy was primarily affected by the fracture dip direction. The quantitative relationship among the connectivity index, permeability coefficient, and permeability anisotropy coefficient was determined based on numerical analysis. The results show that the permeability coefficients in various directions increased with the connectivity index and gradually stabilized. The permeability anisotropy weakened rapidly with increasing connectivity. It stabilized when the connectivity index exceeded 15. The research results have positive significance for revealing the seepage mechanism of complex three-dimensional fracture networks and for preventing and controlling water disasters in fractured rock masses.

1. Introduction

Fractures, such as joints, faults, and bedding planes, are prevalent in crustal rocks. Natural rock masses have undergone long diagenesis and complex geological structural evolution, forming several joints and fissures that significantly change the mechanical properties and seepage characteristics of rock masses and affect the construction safety of underground engineering. Therefore, studying the seepage characteristics of rock-mass fissures is of great theoretical significance and value in engineering applications.

Over the past few decades, numerous scholars have achieved significant research results on the seepage behavior of fractured rock masses through indoor model tests, numerical simulation analyses, and engineering applications. Initially, experiments were conducted on a single smooth fracture, and a cubic law reflecting seepage characteristics was proposed [1,2]. Xu et al. [3] proposed the hypercubic and subcubic laws to reflect the seepage characteristics of rough fractures. With recent advancements in computer technology and numerical calculation meth-

ods, numerical calculations are increasingly applied to rock mechanics and engineering problems. Based on the structure and morphology of a rock mass and several field-measured rock fracture data statistics, the discrete fracture network (DFN) modeling method [4–12], through probability theory and mathematical statistics theory has been applied and verified in several practical projects. The seepage characteristics of complex two-dimensional fracture networks were studied [13–16]. Wang et al. [17] proposed a plane seepage analysis method based on the VC++ software platform to illustrate the permeability tensor characteristics of a two-dimensional fracture network under different geometric distributions. Jafari and Babadagli [18–20] described the connectivity of two-dimensional fracture networks and determined the relationships among the equivalent permeability, fractal dimension, and seepage characteristics.

However, a natural fracture network is distributed in a three-dimensional space, and a two-dimensional fracture network model cannot reflect the actual situation of a naturally fractured rock mass. Therefore, three-dimensional fracture network modeling and seepage

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research are critical issues in this field. Wang et al. [21] combined the line, surface, and body elements into a three-dimensional fracture network model. Zhang et al. [22] and Xie et al. [23] developed a three-dimensional discrete fracture modeling program RJNS^{3D} to model and visualize three-dimensional fracture networks [24–34]. Ivanova et al. [35] summarized the modeling of natural fracture systems at the MIT from 2010 to 2012. They found that the connectivity of three-dimensional fracture networks increased with fracture density and size. In contrast, clustering the fracture directions around a dominant direction reduces connectivity. Subsequently, a numerical study of the seepage characteristics of a three-dimensional fracture network was conducted. Researchers have analyzed seepage characteristics using a mainstream numerical simulation software based on computational fluid dynamics methods, such as ANSYS Fluent [36], OpenFOAM [37], and COMSOL Multiphysics [38]. This commercial software is mature and satisfies most simulation requirements. The available three-dimensional seepage simulation software was further developed to conduct research with the help of C++, Java, MATLAB, etc. Current popular methods include the Galerkin method [39], three-dimensional unified pipe-network method [40], and volume of fluid method [41]. However, the random distribution of fractures in three-dimensional fractured rock masses often has the characteristics of large distribution variability and large numbers. Compared with the two fracture simulation methods, a commercial software often cannot directly model them when the number of fractures is too large. Moreover, there are several challenges in assigning fracture properties and setting boundary conditions; therefore, it is necessary to conduct secondary model development. The self-developed simulation program offers fast calculation and strong convergence but often suffers from low development level and narrow applicability. Additionally, the three-dimensional fracture network structure is complex. Hence, the study of seepage characteristics requires an efficient simulation and accurate connectivity evaluation to reveal the disorder and connectivity of the spatial distribution of the fracture network. However, current research on fracture connectivity evaluation and permeability characteristics treats these aspects relatively independently, and related research still requires improvement.

Therefore, this study proposes a relatively complete technical method for three-dimensional fracture network modeling, fracture connectivity evaluation, and seepage simulation. Using COMSOL Multiphysics integrated with MATLAB, a corresponding calculation program was developed to address complex fracture network modeling, fracture attribute assignment, three-dimensional fracture connectivity evaluation, and a large-scale numerical simulation cycle calculations. Based on this, the connectivity and permeability characteristics of a three-dimensional fracture network were studied, and the influence of fracture geometry parameters on the fracture network connectivity and permeability was explored. The relationship among the connectivity, perme-

ability, and permeability anisotropy of a fractured rock mass was further discussed.

2. Methodology

The model rock mass comprised continuous rock blocks and discontinuous structural planes. The permeability characteristics of the rock blocks are negligible compared to those of the randomly developed fractures in the rock mass. Therefore, the fracture network and connectivity significantly affect the seepage of the fractured rock mass. In actual engineering construction, the distribution of rock mass fractures is often complex, making it challenging to determine the formed fracture network and its connectivity. Therefore, the simulation of rock fracture networks is the primary means of studying the permeability and connectivity of rock fracture networks. First, the Monte Carlo [42] algorithm is employed to generate the spatial distribution of fractures, ensuring a realistic representation of complex fracture networks. The fracture parameters, including the center coordinates, dip angle, dip direction, radius, and aperture, were randomly sampled based on pre-defined probability distributions derived from field data and statistical analysis. To construct the DFN, Monte Carlo sampling is implemented using MATLAB's built-in functions and programming capabilities to generate the necessary dataset, which is subsequently utilized for seepage simulations in COMSOL with MATLAB. The Euler angle was introduced to realize the numerical simulation of complex three-dimensional fracture network models. Finally, the Darcy module of the COMSOL Multiphysics simulation software was used to simulate the seepage of a three-dimensional fracture network and solve the permeability coefficient. In the simulation process, the required parameters are input simultaneously for three-dimensional fracture network modeling and automatic calculation, without manually modifying the parameters, which significantly improves the modeling and calculation efficiency and completes the connectivity and permeability evaluation of the three-dimensional fracture network. This process is illustrated in Fig. 1.

2.1. Generation of a three-dimensional fracture network

During the reconstruction process of a three-dimensional fracture network, the fracture surface is primarily described using the Baecher disc model [43,44]. That is, four essential elements can determine any fracture surface and distribution in space: the position $O(x_0, y_0, z_0)$ of the center point of the fractured disc, fracture occurrence (dip direction α and dip angle β), radius R of the disc, and fracture aperture h , as shown in Fig. 2. The probability distribution models typically used for each element are listed in Table 1.

During modeling, the probability model of the fracture geometry parameters is selected: the center point is generated by a uniform probabil-

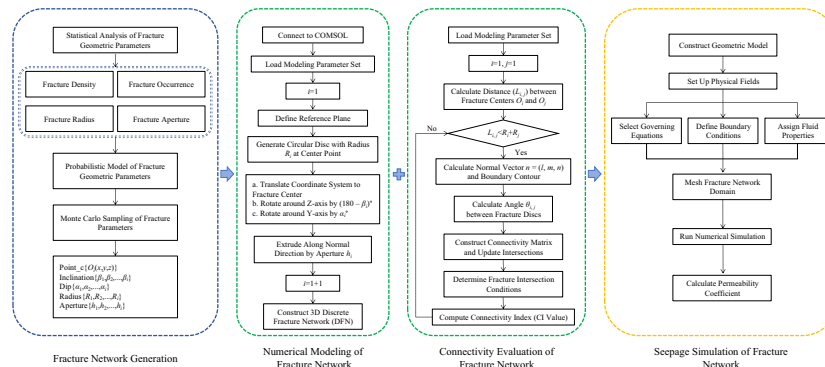


Fig. 1. Schematic diagram of three-dimensional fracture network modeling and evaluation.

Table 1
Probability distribution model of fracture parameters.

Parameter	Distribution	Probability density function
Center point	Poisson distribution	$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad k = 0, 1, 2, \dots$
	Uniform distribution	$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & x < a \text{ or } x > b \end{cases}$
Fracture occurrence	Normal distribution	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(X-\mu)^2}$
	Fisher distribution	$f_{k,m}(t) = \frac{\Gamma(\frac{k+m}{2})}{\Gamma(\frac{k}{2})\Gamma(\frac{m}{2})} t^{\frac{k}{2}-1} (1+t)^{-\frac{k+m}{2}}$
Radius	Gamma distribution	$f(x, \beta, \alpha) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$
	Power-law distribution	$f(x) = x^{-\alpha-1}$
	Lognormal distribution	$p(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$
Fracture aperture	Negative exponential distribution	$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & x < 0 \end{cases}$
	Lognormal distribution	$p(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$

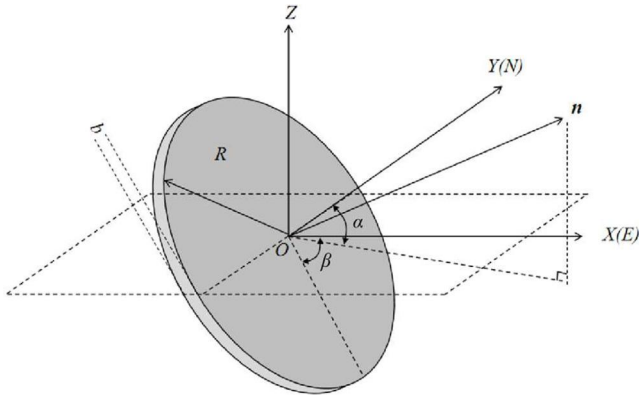


Fig. 2. Diagram of the Baecher disc model.

ity density function, the distribution characteristics of the dip direction and dip angle of the fracture surface obey a normal distribution, the radius of the fracture surface obeys a lognormal distribution [42,45], and the distribution of the fracture aperture obeys a lognormal distribution.

To establish a three-dimensional DFN, the proposed methodology integrates built-in functions and customized algorithms of the MATLAB platform. Following the acquisition of stochastic parameter distributions (e.g., orientation, radius) based on the Baecher disc model, stochastic realizations were generated via Monte Carlo sampling of center coordinates, dip direction (α), dip angle (β), radius (R), and aperture (h). Secondary development was conducted based on COMSOL with the MATLAB module, and the DFN modeling module was established. This platform enables the direct seepage simulation of a three-dimensional DFN by numerically inputting the fracture parameter model (fracture length, dip direction, dip angle, and aperture), boundary conditions, and other parameters.

Coordinate system transformations for individual fracture discs within the COMSOL–MATLAB framework were resolved using Euler angles and rotation matrices. As shown in Fig. 3, it is assumed that the fracture disc parameters are center coordinates $O(x_0, y_0, z_0)$, dip direction α and dip angle β , disc radius R_1 , and fracture width h_1 . First, the fracture disc was drawn according to the center coordinates $O(x_0, y_0, z_0)$ and radius R_1 , and the origin of the global coordinate system was translated to the center point O of the disc. Then, based on the Euler angle theorem, the coordinate system is rotated $(180-\beta)^\circ$ around the z -axis so that the x -axis of the global coordinate system is rotated to the fracture dip direction. Subsequently, the coordinate system is rotated by α around the y -axis so that the x -axis of the global coordinate system coincides with the fracture dip direction, and the fracture disc model is established in the Cartesian coordinate system. According to the size of the fracture aperture h_1 , the fracture disc was stretched symmetrically along the normal direction of the disc to obtain a fracture disc that satisfied the above parameters. Finally, the loop language realizes all fracture disc modeling processes.

By comparing the numerical model built according to the above method with the three-dimensional fracture network image generated by the RJNS^{3D} software, it can be seen that when all fracture parameters are the same, the program can establish the corresponding fracture network model in COMSOL (Fig. 3(c)). Through this method, rapid modeling of the fracture network can be realized in the numerical simulation, and all the fracture parameter information can be obtained directly, which is conducive to the simulation study of the geometric parameter characteristics and seepage characteristics of the fracture network.

2.2. Connectivity characteristics

Rock mass permeability primarily depends on the connectivity of the fracture network. The fracture parameters include fracture density, fracture radius, and fracture occurrence, which are related to the spatial distribution and size of fractures in a three-dimensional space. The

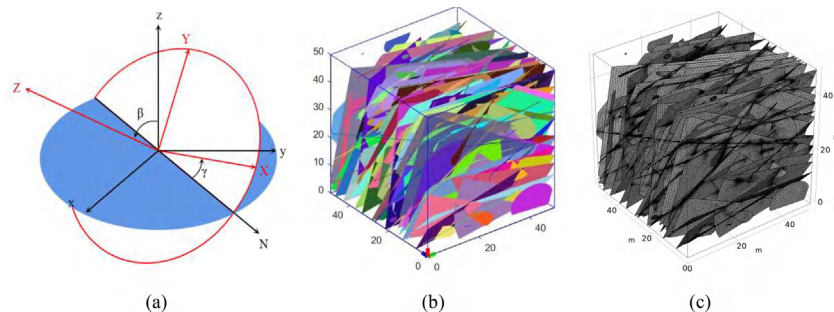


Fig. 3. Schematic diagrams of three-dimensional fracture network modeling. (a) Eulerian angle, (b) RJNS^{3D} software image, and (c) numerical simulation modeling.

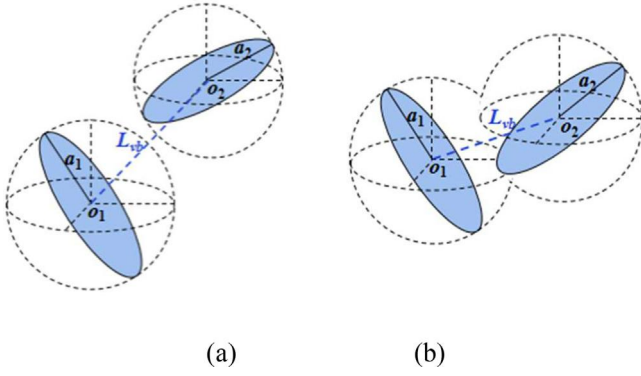


Fig. 4. Schematic diagram of the bounding box method.

spatial distribution characteristics primarily include the relative dimensionless density, fractal dimension, and fracture connectivity. Based on the overall spatial distribution characteristics of the fracture network, these indices focus on revealing the disorder and connectivity of its spatial distribution of the fracture network.

In the seepage process of fractured rock masses, it is challenging for isolated fractures or fractures with poor connectivity to contribute substantially to fluid migration. Therefore, the fluid flow in the fracture network primarily depends on the connectivity of the rock fracture network. The number of fracture intersections and fracture discs characterizes fracture connectivity. The connectivity index CI [18,20] is defined as follows:

$$CI = \frac{M}{N} \quad (1)$$

In a three-dimensional study area, M is the number of intersecting lines between the fracture discs in the region, and N is the number of fracture discs within the region.

The intersection frequency of the fracture disks was accurately calculated to obtain the connectivity of the three-dimensional fractured rock mass. Based on the research methodology provided in [13], a MATLAB program was developed to determine the intersection of fracture discs. First, the bounding box method preliminarily determines the spatial relationship between two cracks. Fig. 4 shows that the overlap of the two fractures surrounding the sphere can be analyzed by comparing the linear distance L_{bv} between the centers of the two bounding spheres and the radii of the spheres. The geometric characteristics are as follows:

$$L_{bv} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \leq (a_1 + a_2) \quad (2)$$

where L_{bv} is the distance between the spheres surrounding the two discs, $x_1, y_1, z_1, x_2, y_2, z_2$ are the global coordinates of the center of the two discs, and a_1 and a_2 represent the radii of the corresponding fracture discs.

When $L_{bv} < (a_1 + a_2)$, the positional relationship between the two fractures was calculated. According to the occurrence of the fracture disc (dip direction α and dip angle β), the average vector of the fracture disc $\mathbf{n} = (l, m, n)$ can be expressed as:

$$\begin{cases} l = \sin \beta \sin \alpha \\ m = \sin \beta \cos \alpha \\ n = \cos \beta \end{cases} \quad (3)$$

The plane of the first fracture disc can be expressed as:

$$l_1(x - x_1) + m_1(y - y_1) + n_1(z - z_1) = 0 \quad (4)$$

The boundary contour of the first fracture disc can be expressed as:

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = a_1^2 \quad (5)$$

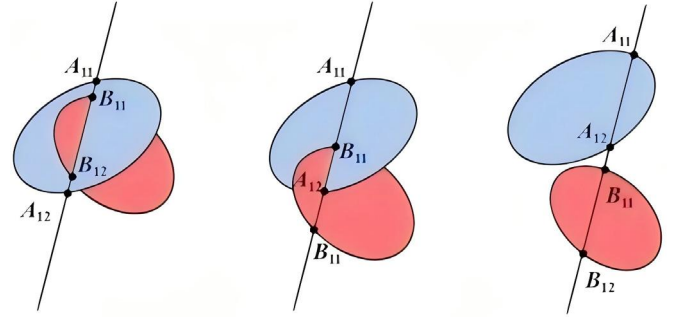


Fig. 5. Schematic diagram of the fracture intersection. (a) Inclusion, (b) intersection, and (c) displacement.

Similarly, the plane and boundary contour lines of the second fracture disc can be expressed as:

$$l_2(x - x_2) + m_2(y - y_2) + n_2(z - z_2) = 0 \quad (6)$$

$$(x - x_2)^2 + (y - y_2)^2 + (z - z_2)^2 = a_2^2 \quad (7)$$

When the two surrounding spheres overlap, the angle θ between the planes of the two fracture discs should be calculated first, which can be determined by the following formula:

$$\theta = \cos^{-1} \frac{|l_1 l_2 + m_1 m_2 + n_1 n_2|}{\sqrt{(l_1^2 + m_1^2 + n_1^2) \cdot (l_2^2 + m_2^2 + n_2^2)}} \quad (8)$$

When $\theta = 0$, the plane of the two fracture discs is parallel, and the intersection line of the fracture discs does not exist. In contrast, the above formula can calculate whether the two fracture discs intersect at the intersection line. When the two fissure discs intersect with the intersection line, the four space intersection points of the two fissure discs and the intersection line should be used to analyze and judge the positional relationship of the fissure. As shown in Fig. 5(a) and (b), the second fracture is included in or intersects the first fracture, indicating an intersecting relationship between the two fracture discs. Simultaneously, the spatial equation of the intersecting line and intersecting length can be obtained using four coordinates. Fig. 5(c) shows that the two fractures are separated, indicating that although the two fracture discs have a unified intersection line, they do not intersect.

To calculate the connectivity of the fracture network, according to the bounding box method, the intersection between two fracture disks can be solved and judged, the connection matrix established, and the number of fracture intersections updated. Finally, the CI value of the fracture network was calculated using a conditional cyclic language, and the connectivity evaluation of the fracture network was completed.

2.3. Fluid flow through a three-dimensional fracture network

2.3.1. Governing equations and boundary conditions

This simulation considered the groundwater flow in three directions— x , y , and z . The inlet and outlet boundaries of the model were set as water pressure boundaries, where the pressure at the inlet was 10,000 Pa, the pressure at the outlet was 0 Pa, and the remaining boundaries were impervious.

The permeability of the fracture network can be calculated by modeling the incompressible steady seepage of the connected network, as shown in Fig. 6. By generalizing Darcy's law, the seepage velocity within the fracture element ij in local coordinates can be expressed as

$$u = -\frac{\gamma_w k_f}{\mu} \frac{\partial H}{\partial l} \quad (9)$$

where u is the velocity of fluid flow, μ is the dynamic viscosity of water, γ_w ($\gamma_w = \rho g$) is the unit weight of water, k_f is the fracture permeability, and $\partial H / \partial l$ is the hydraulic head drop. The correlation between the

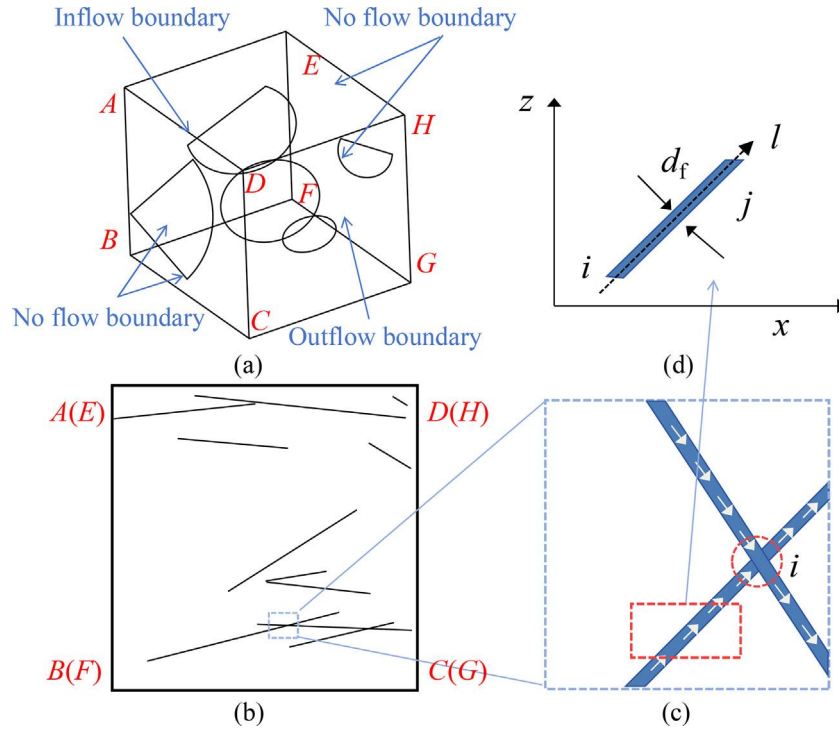


Fig. 6. (a) Illustration of a three-dimensional DFN with a set of N grids, (b) section diagram of the seepage path, (c) mass conservation at the fracture intersection, and (d) smooth parallel-plate model in the local coordinate system.

hydraulic pressure drop $\partial P/\partial l$ and hydraulic head drop $\partial H/\partial l$ is:

$$\frac{\partial H}{\partial l} = \frac{\partial P}{\partial l \cdot \gamma_w} \quad (10)$$

The cubic law is [46,47]:

$$k_f = \frac{d_f^2}{12f_f} \quad (11)$$

where d_f is the hydraulic aperture, and f_f is the roughness coefficient.

The governing equation of a single fracture in Fig. 6(d) can be expressed as [45]:

$$\frac{\partial \mathbf{u}}{\partial l} = 0 \quad (12)$$

where $\mathbf{u}$ is the fluid velocity vector.

As shown in Fig. 6(c), there are m_i connected fracture units at the intersection i , which cannot store water under stable flow conditions. Therefore, the sum of the inflowing and outflowing fracture flows should remain constant. The law of conservation of the inflow and outflow velocities at intersection i is expressed as follows:

$$\sum_{j=1}^{m_i} d_{f_j} \mathbf{u}_j = 0 \quad (13)$$

To solve the equation for an entire fracture network [18,20], the flow must satisfy the following boundary conditions:

(1) The water head boundary conditions:

$$P_i = P_1 - P_2 \quad (14)$$

where the inflow boundary $ABFE$ was P_1 , and the outflow boundary $DCGH$ was P_2 .

(2) The flux boundary conditions:

$$q_i = d_f \mathbf{u}_i = 0 \quad (15)$$

The $ABEF$, $BCGF$, $EFGH$, and $ADHE$ boundaries are impermeable.

2.3.2. Numerical solution procedure

The fracture surface was divided into triangular meshes, and the flow and pressure fields were solved. The steady-state volume flow velocity was obtained by integrating the flow velocity at the entrance for different fracture aperture distributions. Darcy's law is used to determine the permeability coefficient of the fracture network. The specific process is as follows:

$$K = \frac{QL}{A(P_1 - P_2)} \quad (16)$$

where Q is the volume flow, A is the cross-sectional area, L is the length of the fracture network in the flow direction, and $P_1 - P_2$ are the head pressure differences in the cross section. The flow section A at the outlet can be obtained by

$$A = \int_{i=1}^n L_i b_i \quad (17)$$

The flow velocity Q of the cross section is obtained using Eq. (18).

$$Q = \int_{i=1}^n A_i u_i \quad (18)$$

2.4. Fissure network model generation

COMSOL Multiphysics is a cross-platform finite-element analysis, solution, and multiphysics simulation software. This provides a coupling system for the traditional physics-based user interfaces and partial differential equations. In addition to providing built-in modules to solve problems, it can set model parameter variables using external control software.

This study developed a joint simulation program using COMSOL and MATLAB. The advantages of the two software programs were combined to realize three-dimensional DFN modeling and connectivity evaluation. For simplicity, each fracture is assumed a smooth parallel plate, with its other parameters described using probability density functions. The center point is generated by a uniform probability density function, and the distribution characteristics of the dip direction and dip angle of the

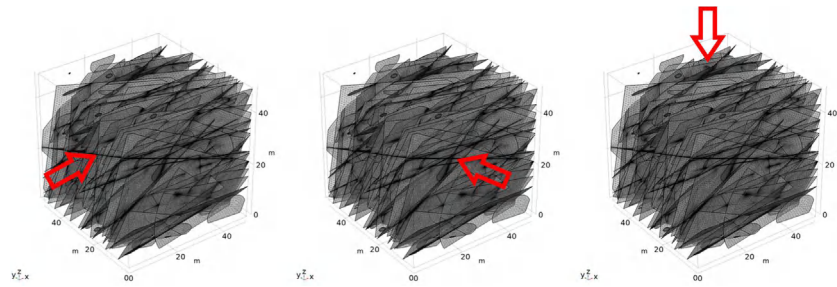


Fig. 7. Numerical model diagrams of different seepage directions.

fracture surface obey a normal distribution. The radius of the fracture surface and the fracture aperture followed lognormal distributions. The mean value was set to 1 mm.

To estimate the influence of the fracture geometry parameters on the fracture network connectivity and permeability characteristics, multiple three-dimensional fracture networks were generated in a $50 \times 50 \times 50$ m³ area. Meanwhile, to eliminate the influence of the size effect on the numerical simulation results, the crack length L_f was controlled in the simulation process, and the appropriate crack length was selected by introducing the dimensionless parameter $D = L_f/L_m$ [48–50]. The dimensionless parameters D selected in this study were 0.2, 0.3, 0.4, and 0.5. The fracture lengths were 10, 15, 20, and 25 m. The pressure head difference ΔP between each fracture network's inflow and outflow boundaries was 10,000 Pa. Fig. 7 shows three examples of three-dimensional fracture networks with various seepage directions.

The physical field was set after establishing the three-dimensional fracture geometric model. First, the control equation was selected, the boundary conditions and fluid parameters were set, the density and dynamic viscosity were defined, and the permeability coefficient was solved by dividing the fracture grid and running the simulation.

Meshing in COMSOL can be performed by selecting the existing grid options. The finer the grid, the more accurate the numerical simulation results; however, finer grids significantly increased the calculation time. Grid independence was analyzed to reduce the numerical solution time and obtain the exact solution. The 10-1 model was selected for verification. The numerical results of the extremely refined, ultra-refined, relatively refined, refined, conventional, coarse, and relatively coarse mesh generations were analyzed. The results, depicted in Fig. 8, indicate a numerical difference in the conventional, refined, coarse, and relatively coarse permeable data. However, the data between the ultra-refined, extremely refined, and relatively refined grids showed almost no difference, and the relatively refined grids saved considerable calculation time compared to the extremely refined grids. Therefore, to balance accuracy and computing cost, relatively refined grids were chosen to divide the physical field.

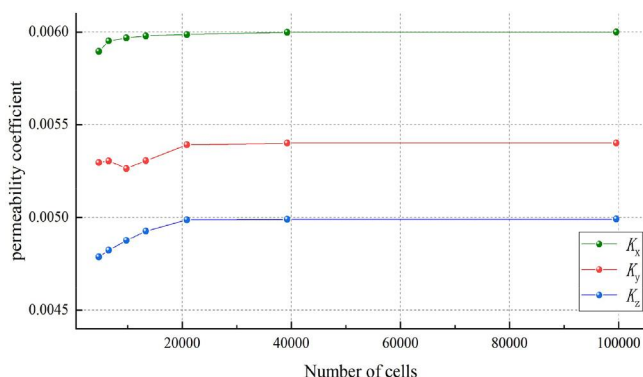


Fig. 8. Permeability experiment results across various meshing methods.

Table 2

Test values for the four levels of influencing factors considered in the study.

Levels/factors	$\rho/\text{strip}\cdot\text{m}^{-1}$	D	L/m	$\alpha/^\circ$	$\beta/^\circ$
1	0.0004	0.2	10	0	20
2	0.0008	0.3	15	25	40
3	0.0024	0.4	20	50	60
4	0.0032	0.5	25	75	80

3. Results and analysis

3.1. Scenarios of three-dimensional synthetic fracture networks

To comprehensively study the influence of fracture geometry parameters on the connectivity and seepage characteristics of the fracture network, fracture density ρ , fracture dip direction α , fracture dip angle β , and fracture length L were selected as the experimental factors in this study. The connectivity and equivalent permeability coefficients of different fracture occurrence combinations were analyzed using orthogonal numerical tests. A four-factor and four-level orthogonal test table, $L_{16} (4^4)$, was selected. The specific parameters are listed in Table 2. Based on this combination, the fractures were divided into three types: type I, one group of fractures; type II, two groups of fractures; and type III, three groups of fractures. The specific fracture parameters are listed in Table 3.

The orthogonal test in this study adopted a balanced design to ensure a systematic and controlled evaluation of the influence of geometric fracture parameters on the connectivity and permeability of the fracture networks. Specifically, the number of fractures in types II and III was uniformly distributed among different groups. This setting, although simplified than the heterogeneous distribution in natural rock masses, enables the isolation and quantification of the effects of individual parameters by reducing confounding influences. This controlled design facilitates statistical analysis and highlights the relative importance of the critical factors. Furthermore, the dip direction and angle were discretized at 60° intervals. This selection captures representative fracture orientations, namely horizontal (0°), intermediate (60°), and steep (120° or 180°), which are typically observed in geological formations. This choice strikes a balance between geometric diversity and computational tractability, allowing for efficient modeling while preserving meaningful variations in orientation. These simplifications may limit the generalizability of the results.

As shown in Fig. 9, 48 fracture network models were generated based on the specific parameters of the orthogonal test fracture model and fracture network generation model. For example, the label “1-1” in the figure represents the first fracture combination type under the orthogonal test Scheme 1. Different fracture generation parameters result in variations in the fracture network density and fracture distribution direction. Moreover, even with the same fracture parameters, different fracture combinations affected the spatial distribution of the fracture network. Forty-eight fracture network simulations were conducted across the three seepage directions, resulting in 144 simulations. This compre-

Table 3
Specific parameters of the different fracture combination types considered in the study.

Fracture type	Fracture density	Fracture dip direction	Fracture dip angle	Fracture length
I	ρ	α	β	L
II	$\rho/2$	α	β	L
	$\rho/2$	$\alpha + 60^\circ$	β	L
III	$\rho/3$	α	β	L
	$\rho/3$	$\alpha + 60^\circ$	β	L
	$\rho/3$	random	β	L

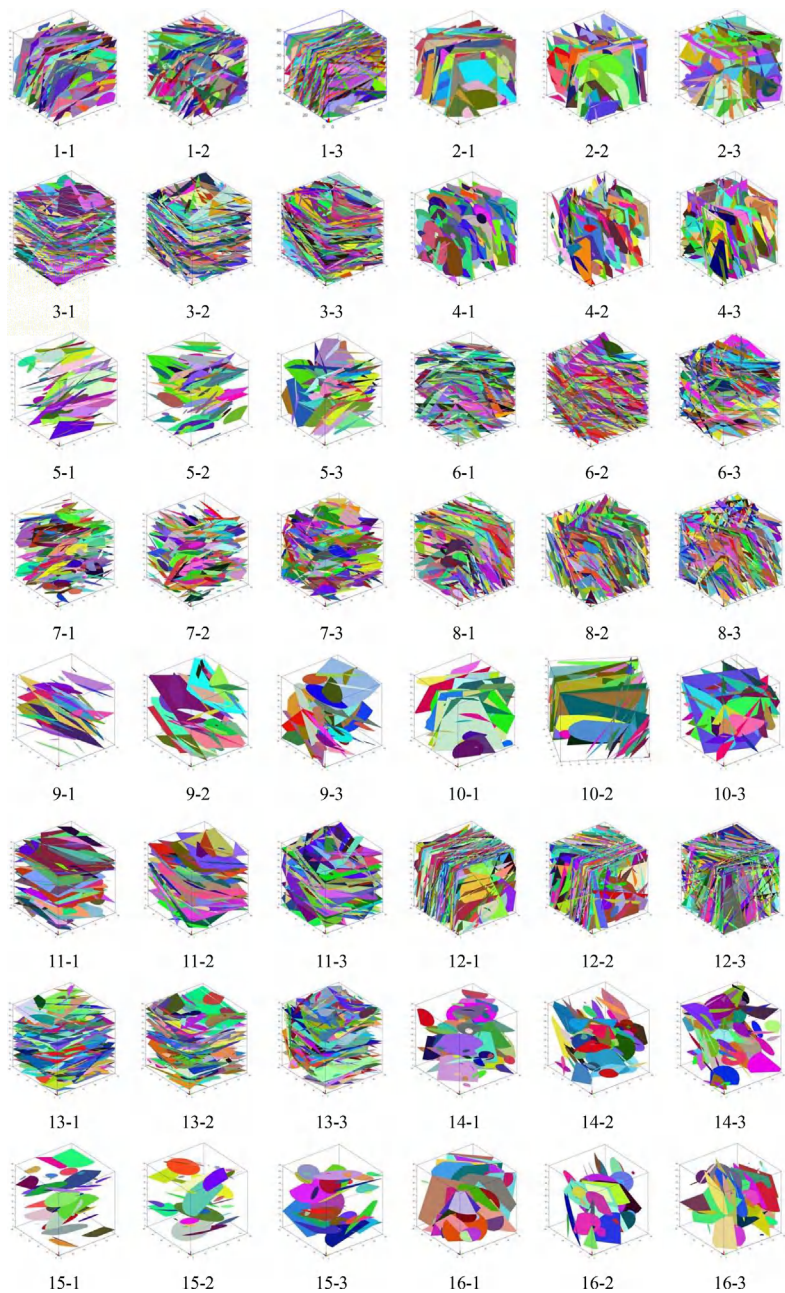


Fig. 9. Schematic diagram of the fracture network model.

hensive approach enhances the representation of the actual rock fracture seepage behavior under varying degrees of heterogeneity.

3.2. Connectivity characteristics of the fracture network

The fracture network *CI* was selected as the connectivity index, and the fracture network connectivity evaluation module calculated the *CI*

to evaluate the influence of each fracture parameter on the connectivity of the fracture network. The specific test scheme and calculation results are presented in Table 4.

(1) Analysis of the factors influencing connectivity

The orthogonal test results in this study were analyzed using variance analysis. Variance analysis is a mathematical method that separates the differences in test results caused by different factors and levels from the

Table 4

Parameter modeling and simulation results of the fracture network connectivity and permeability coefficient.

Trial.	$\rho/\text{strip}\cdot\text{m}^{-3}$	L/m	$\alpha/^\circ$	Type	Number of intersecting fractures	CI	$Kx/\text{cm}\cdot\text{s}^{-1}$	$Ky/\text{cm}\cdot\text{s}^{-1}$	$Kz/\text{cm}\cdot\text{s}^{-1}$	Ckx/y	Ckx/z	Cky/z
1-1	0.0024	20	0	I	3957	13.19	0.016	0.003	0.017	5.534	0.933	0.169
1-2	0.0024	20	0	II	3981	13.27	0.011	0.014	0.012	0.828	0.982	1.186
1-3	0.0024	20	0	III	3606	12.02	0.012	0.011	0.016	1.185	0.759	0.641
2-1	0.0008	20	25	I	881	8.81	0.010	0.004	0.012	2.290	0.827	0.361
2-2	0.0008	20	25	II	611	6.11	0.005	0.018	0.016	0.263	0.302	1.149
2-3	0.0008	20	25	III	870	8.7	0.010	0.012	0.012	0.851	0.873	1.025
3-1	0.0032	20	50	I	8480	21.2	0.011	0.013	0.007	0.829	1.524	1.839
3-2	0.0032	20	50	II	8932	22.33	0.011	0.013		0.891		
3-3	0.0032	20	50	III	14212	35.53	0.010	0.011		0.897		
4-1	0.0024	10	75	I	1146	3.82	0.000	0.005	0.006	0.107	0.082	0.771
4-2	0.0024	10	75	II	1107	3.69	0.001	0.004	0.005	0.338	0.264	0.782
4-3	0.0024	10	75	III	1536	5.12	0.005	0.006	0.009	0.843	0.628	0.745
5-1	0.0008	15	0	I	470	4.7	0.014	0.004	0.004	3.594	4.000	1.113
5-2	0.0008	15	0	II	311	3.11	0.008	0.005	0.001	1.429	9.823	6.873
5-3	0.0008	15	0	III	529	5.29	0.009	0.006	0.004	1.698	2.638	1.554
6-1	0.0024	25	50	I	4755	15.85	0.008	0.008	0.005	0.913	1.649	1.806
6-2	0.0024	25	50	II	10704	35.68	0.062	0.068		0.912		
6-3	0.0024	25	50	III	7926	26.42	0.012	0.013	0.016	0.918	0.721	0.785
7-1	0.0032	10	25	I	1876	4.69	0.007	0.002	0.002	2.788	3.068	1.100
7-2	0.0032	10	25	II	2380	5.95	0.007	0.006	0.009	1.342	0.810	0.604
7-3	0.0032	10	25	III	3312	8.28	0.006	0.013	0.010	0.451	0.602	1.336
8-1	0.0032	15	75	I	6416	16.04	0.002	0.024	0.011	0.068	0.153	2.265
8-2	0.0032	15	75	II	6628	16.57	0.007	0.010	0.016	0.662	0.422	0.638
8-3	0.0032	15	75	III	7392	18.48	0.005	0.010	0.013	0.498	0.368	0.738
9-1	0.0004	20	75	I	134	2.68	0.001	0.005	0.004	0.164	0.187	1.142
9-2	0.0004	20	75	II	182	3.64	0.006	0.006	0.003	1.146	2.013	1.756
9-3	0.0004	20	75	III	109	2.18	0.007	0.014	0.008	0.473	0.842	1.781
10-1	0.0004	25	25	I	119	2.38	0.006	0.005	0.005	1.111	1.205	1.084
10-2	0.0004	25	25	II	385	7.7	0.004	0.002	0.008	2.080	0.529	0.254
10-3	0.0004	25	25	III	185	3.7	0.005	0.009	0.017	0.507	0.287	0.566
11-1	0.0008	25	75	I	839	8.39	0.000	0.012	0.002	0.001	0.006	6.783
11-2	0.0008	25	75	II	1084	10.84	0.017	0.019	0.004	0.922	3.998	4.337
11-3	0.0008	25	75	III	1194	11.94	0.012	0.013	0.005	0.984	2.395	2.434
12-1	0.0032	25	0	I	10360	25.9	0.026	0.009		2.956		
12-2	0.0032	25	0	II	10532	26.33	0.011	0.013		0.820		
12-3	0.0032	25	0	III	11768	29.42	0.029	0.092		0.317		
13-1	0.0024	15	25	I	3633	12.11	0.013	0.012	0.003	1.085	4.829	4.450
13-2	0.0024	15	25	II	3747	12.49	0.013	0.014	0.007	0.953	1.741	1.827
13-3	0.0024	15	25	III	3486	11.62	0.020	0.016	0.010	1.254	1.935	1.543
14-1	0.0008	10	50	I	113	1.13	0.000		0.000		0.845	
14-2	0.0008	10	50	II	187	1.87	0.002	0.005	0.003	0.356	0.582	1.635
14-3	0.0008	10	50	III	143	1.43	0.002	0.005	0.004	0.383	0.499	1.303
15-1	0.0004	10	0	I	58	1.16	0.003	0.003	0.002	1.213	1.715	1.413
15-2	0.0004	10	0	II	40	0.8	0.001		0.000		1.333	
15-3	0.0004	10	0	III	50	1	0.002	0.001	0.001	1.762	3.167	1.798
16-1	0.0004	15	50	I	148	2.96	0.000	0.002	0.008	0.192	0.037	0.191
16-2	0.0004	15	50	II	88	1.76	0.000	0.004	0.002	0.007	0.013	1.800
16-3	0.0004	15	50	III	102	2.04	0.005	0.007	0.003	0.664	1.622	2.444

Table 5

Analysis of variance results of connectivity analysis.

Source of deviation	Square of deviance	Degree of freedom	Mean square deviation	Monitor value F	Significance P
$\rho/\text{strip}\cdot\text{m}^{-1}$	2018.613	3	672.871	45.811	0
L/m	1221.71	3	407.237	27.726	0
$\alpha/^\circ$	292.673	3	97.558	6.642	0.001
$\beta/^\circ$	77.926	3	25.975	1.768	0.172
N	48.206	2	24.103	1.641	0.209
Error	484.699	33	14.688		
grand total	9358.829	48			
total adjusted error	4143.827	47			

differences in effects caused by error fluctuations. This method decomposes the total deviation square sum of values into the deviation square sum caused by factors and the deviation square sum caused by errors, calculates the average deviation square sum, constructs F statistics, and determines the significance of each element. A higher F -value indicated a stronger influence of the corresponding variable on the results. The results are summarized in Table 5.

According to the results of variance analysis, the significance of the influence on connectivity from large to small is as follows: ρ ($F = 45.811$)

$> L$ ($F = 27.726$) $> \alpha$ ($F = 6.642$) $> \beta$ ($F = 1.768$) $> N$ ($F = 1.641$). According to the significance value P , when $0.01 < P < 0.05$, the difference is substantial; when $P < 0.01$, the difference is significant; and when $P > 0.05$, the difference is insignificant. The analysis results showed that the fracture density, fracture length, and fracture dip direction significance values P were less than 0.01. These three parameters had a considerable influence on the connectivity of the fracture network, while the fracture dip angle and fracture group number had no significant impact on connectivity.

(2) Fracture network connectivity estimation

Based on the variance analysis, the influence of the geometric parameters on the connectivity of the fracture network was obtained. However, in an actual rock fracture network, because of the significant number of fractures and complex interlacing, it is impossible to calculate the number of intersecting fractures in the network and obtain an accurate CI value. Therefore, an estimation model of the geometric parameters and connectivity of the fracture network was established using the calculation results of an orthogonal test, which is conducive to the rapid characterization of connectivity and the guidance of production practice. A multivariate linear regression model was used for the analysis. Assuming that the fracture geometry parameters are related to connectivity, the multivariate linear regression model can be expressed as:

$$CI = a\rho + bL + c\alpha + d\beta + eN + f \quad (19)$$

where a , b , c , d , and e are regression coefficients, and f is a constant.

In this study, the SPSS numerical analysis software was used to establish a multiple linear regression model using the least squares method to determine the regression coefficients a , b , c , d , e , and f . After calculation, the fracture network connectivity estimation model was expressed as follows:

$$CI = 5647.074\rho + 0.899L - 0.035\alpha - 0.008\beta + 1.193N - 15.238 \quad (20)$$

The goodness of fit of the model was $R^2 = 0.795$, indicating that the established CI value estimation model could better estimate the connectivity of the fracture network.

3.3. Relationship between fracture connectivity, permeability, and permeability anisotropy

Fig. 10(a)–(i) show the distributions of the Darcy velocity field in the x , y , and z directions of the three-dimensional fracture network in

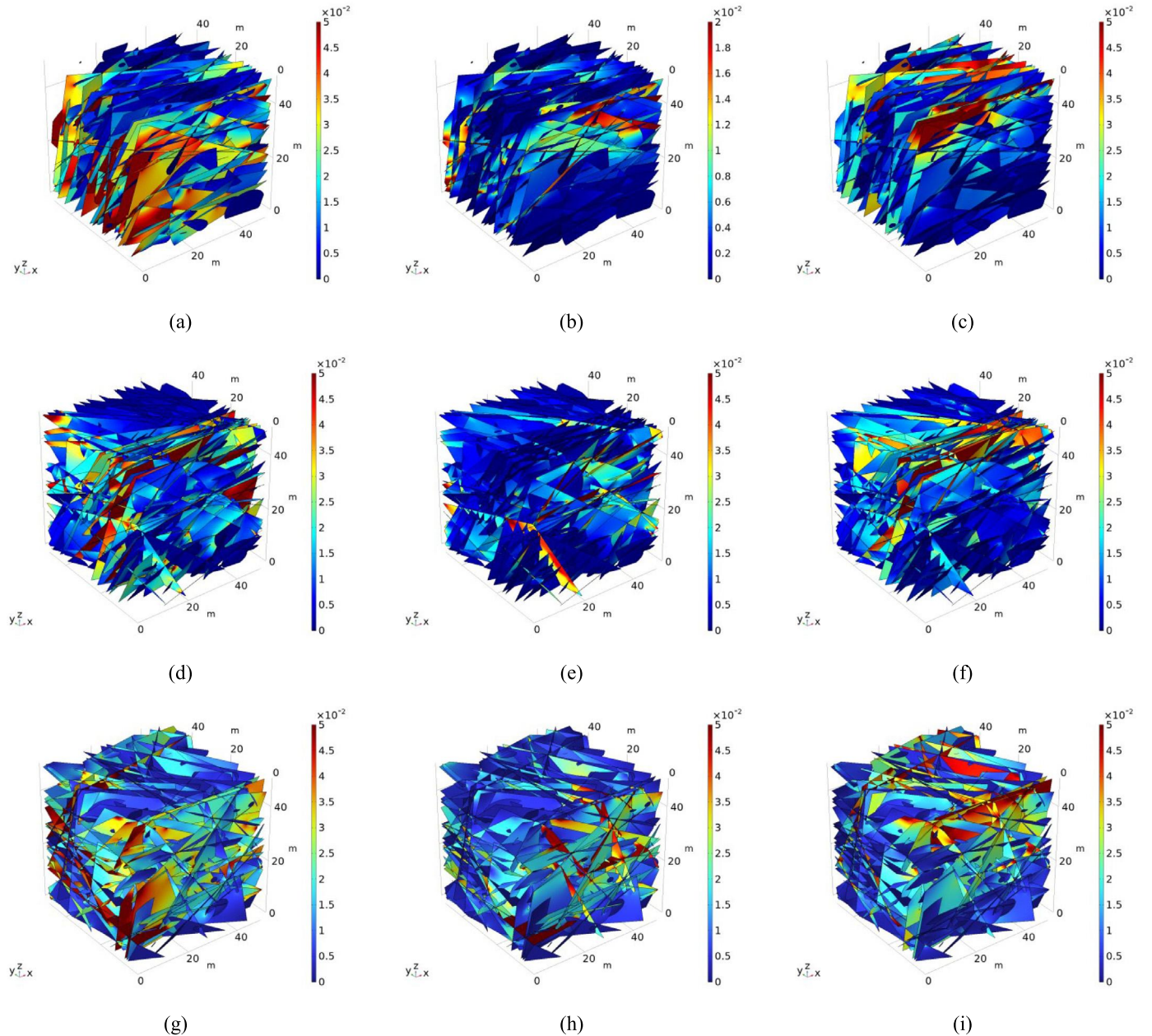


Fig. 10. Darcy velocity field distribution of the fracture network under various test schemes and seepage directions. (a) Test 1-1 x direction, (b) test 1-1 y direction, (c) test 1-1 z direction, (d) test 2-1 x direction, (e) test 2-1 y direction, (f) test 2-1 z direction, (g) test 3-1 x direction, (h) test 3-1 y direction, and (i) test 3-1 z direction.

Table 6
Results of variance analysis of the permeability coefficient.

Source of deviation	Square of deviance	Degree of freedom	Mean square deviation	Monitor value F	Significance P
$\rho/\text{strip}\cdot\text{m}^{-1}$	0	3	0	7.512	0.001
L/m	0	3	0	8.093	0
$\alpha/^\circ$	0	3	0	6.151	0.002
$\beta/^\circ$	8.13E-05	3	2.71E-05	1.465	0.242
N	6.24E-05	2	3.12E-05	1.687	0.201
Error	0.001	33	1.85E-05		
Grand total	0.005	48			
Total adjusted error	0.002	47			

Test 1. The color of the figure, from blue to red, reflects the fluid velocity inside the fracture network, from slow to fast. When the color was dark blue, the fluid rate was 0 m/s, indicating that the fracture was isolated and disconnected from other fractures. Fig. 10(a)–(c) show that the distribution of the Darcy velocity field is significantly different in various seepage directions under the same fracture parameters and combination types. The seepage process was evident when the seepage inlet was perpendicular to the direction of the fracture dip. When the seepage inlet was parallel to the fracture dip direction, the fluid flow was slow and exhibited significant directivity. Simultaneously, using Eq. (12), the permeability coefficients in the x , y , and z directions are 0.01566, 0.00283, and 0.01679 cm/s, respectively. The anisotropy of the fracture significantly affects the seepage of the rock mass. Fig. 10 (a), (d), and (g) show the x -direction seepage under the same fracture parameters but different combination types. The Darcy velocity field distributions of the fracture networks under different combinations are also shown. In Fig. 10(a), the fracture seepage channel is dominated by fractures parallel to the inlet direction, and the fluid velocity is high, as shown in Fig. 10(d) and (g). With the increase in the number of fracture groups and the disordered change in some fractures, the fluid begins to flow throughout the fracture network. Nevertheless, the fluid velocity also decreased owing to a shift in the path. By observing the permeability coefficient of the fracture network under different fracture combinations, it is found that the permeability coefficients for the three combination types are 0.01566, 0.01148, and 0.01244 cm/s. The permeability coefficient difference was insignificant, indicating that the fracture seepage process was primarily affected by the fracture geometry parameters.

Combined with the simulation data in Table 4, the permeability coefficient in the x -direction was analyzed using variance analysis. The calculation results are listed in Table 6. The results of variance analysis show that the influence of fracture geometry parameters on permeability is as follows: L ($F = 8.093$) > ρ ($F = 7.512$) > α ($F = 6.151$) > N ($F = 1.465$) > β ($F = 1.687$). The fracture length, density, and dip direction significantly affected the permeability of the fracture network. These results support those of Gao et al. [51], who emphasized the influence of fracture parameters on the fracture network permeability using two-dimensional fracture network modeling.

Fig. 11 shows the relationship between the fracture network connectivity and permeability coefficient. The analysis showed a positive correlation between the CI value of the fracture network connectivity and the permeability coefficient in different directions. The permeability coefficient of the fracture network increased with increasing fracture connectivity.

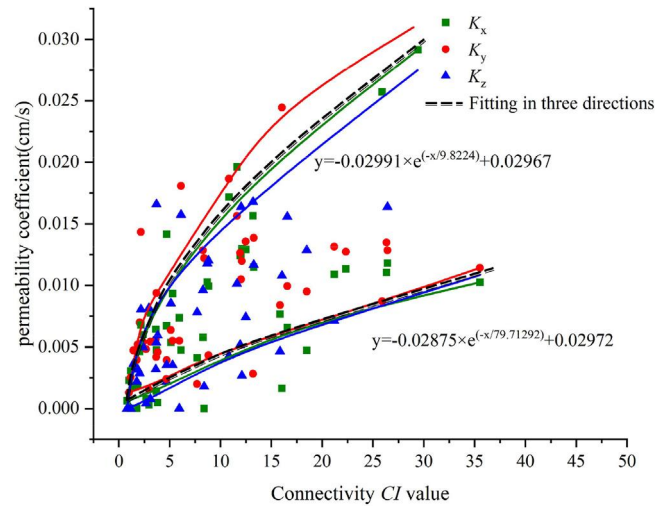


Fig. 11. Relationship between the fracture network connectivity and permeability coefficient.

cent of the fracture network increased with increasing fracture connectivity. Fitting analysis shows that a logarithmic function better describes the relationship between the two, and the similar positions corresponding to the seepage models in different directions indicate that the relationship between the CI value and permeability coefficient remains consistent. At low CI values, the permeability coefficient increased with the CI value; however, when the CI value reached a specific value, the permeability coefficient tended to stabilize.

To analyze the influence of the fracture geometry parameters on the permeability anisotropy of a fracture network, the permeability coefficient ratios K_x/K_y , K_x/K_z , and K_y/K_z in different directions were used as evaluation indices, which are collectively referred to as the permeability anisotropy coefficient C_k . Clearly, a larger C_k indicates a more evident permeability anisotropy in the fracture network. Combined with the data in Table 4, K_x/K_y was analyzed using variance analysis. The calculation results are listed in Table 7. According to the study of variance for fracture anisotropy, the significance of fracture geometry parameters is as follows: α ($F = 5.776$) > N ($F = 2.188$) > β ($F = 0.644$) > L

Table 7
Variance results of the anisotropic coefficient.

Source of deviation	Square of deviance	Degree of freedom	Mean square deviation	Monitor value F	Significance P
$\rho/\text{strip}\cdot\text{m}^{-1}$	1.372	3	0.457	0.562	0.644
L/m	1.439	3	0.48	0.589	0.626
$\alpha/^\circ$	14.096	3	4.699	5.776	0.003
$\beta/^\circ$	1.572	3	0.524	0.644	0.592
N	3.559	2	1.78	2.188	0.128
error	26.847	33	0.814		
grand total	101.545	48			
total adjusted error	48.885	47			

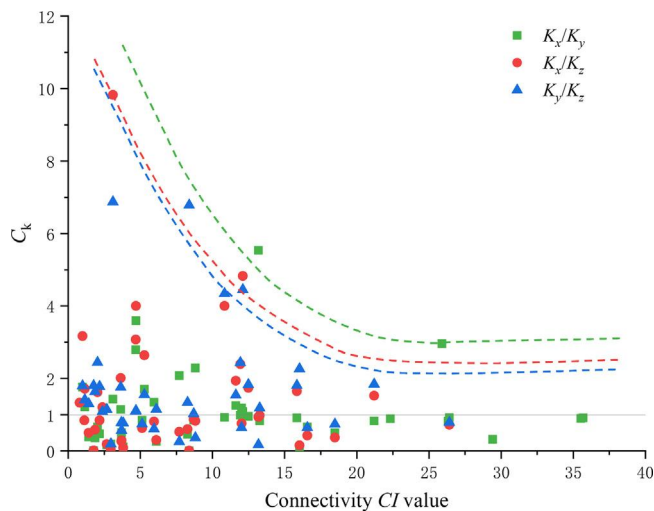


Fig. 12. Relationship between the fracture network connectivity and anisotropy coefficient.

($F = 0.562$) > ρ ($F = 0.589$). Combined with the significant P value, the anisotropy effect of the fracture dip direction on the permeability of the fracture network was extremely significant, and the anisotropy effect of other parameters on the fracture network was ineffective.

Furthermore, Fig. 12 shows that the relationship between the connectivity CI value and anisotropy coefficient C_k in various directions are the same. When the connectivity of the fracture network is low, especially when $CI < 15$, the seepage path is less and relatively fixed, resulting in a large range of anisotropy coefficient C_k . As connectivity increased and exceeded $CI > 15$, the value of C_k became more stable. This indicates that enhanced fracture network connectivity reduces the permeability anisotropy induced by the rock mass structure. In practical engineering applications, the permeability coefficient of a rock mass can be preliminarily estimated based on the connectivity index.

4. Conclusion

This study developed a comprehensive methodology for modeling, evaluating, and simulating the permeability and connectivity of three-dimensional DFNs utilizing COMSOL coupled with a MATLAB secondary development platform. This approach enables rapid DFN construction and permeability analysis, providing a practical solution for large-scale fractured rock mass simulation. According to the numerical simulation results, the following conclusions are drawn:

- (1) A bounding box algorithm was employed to determine the fracture intersections and calculate the CI of 3D fracture networks. Based on the numerical simulation results, a multivariate linear regression equation was constructed to predict CI from the geometric fracture parameters.
- (2) Through an orthogonal experimental design, the influence of fracture geometry (including fracture density, length, dip direction, dip angle, and combination type) on the fracture network connectivity, permeability, and permeability anisotropy was systematically analyzed. The fracture density, length, dip direction, dip angle, and combination type significantly affected connectivity. Permeability is primarily influenced by the fracture length and density, followed by the dip direction and combination type. The permeability anisotropy was mostly affected by the fracture orientation parameters.
- (3) The simulation revealed clear correlations between the CI , permeability coefficient (K), and permeability anisotropy coefficient (C_k). As connectivity increased, permeability increased and tended to stabilize, whereas anisotropy decreased. When $CI > 15$,

C_k approaches stability, indicating that the enhanced connectivity weakens the anisotropic behavior of the fractured media.

Overall, this study offers new insights into the relationship between fracture geometry, connectivity, and flow behavior and provides a practical methodology for the large-scale simulation and prediction of fractured rock mass permeability. Future work will further consider the variability in the fracture aperture and roughness, as well as broader parameter ranges, to enhance the applicability and generalizability of the proposed approach.

5. Discussion and limitations

This paper presents a numerical framework for modeling and evaluating the connectivity and permeability of three-dimensional DFNs. Compared with traditional two-dimensional models, the three-dimensional method developed in this study more accurately captures the spatial heterogeneity and anisotropic permeability of natural rock masses, offering a new perspective for DFN-based permeability analysis. Additionally, the customized simulation program developed using COMSOL and MATLAB significantly reduced the computational costs, improved the modeling efficiency, and enhanced its applicability in underground reservoir design, hydraulic engineering, and geological storage systems.

An orthogonal experimental design was used to generate 48 DFN realizations for the regression analysis, resulting in a multivariate linear regression model with an R^2 value of 0.795, indicating a moderate fit. Although the model serves as a useful preliminary tool for estimating connectivity, the limited number of realizations affects its robustness and generalizability. Future work will increase the number of realizations to improve the statistical strength of the analysis; however, this model has some limitations. This does not consider the randomness of the fracture aperture and surface roughness, which significantly affects flow behavior and connectivity. Additionally, the parameter ranges, particularly for fracture orientations (dip angle α and dip direction β), were limited to discrete values, restricting the model's ability to represent the full variability of natural fracture systems.

Future research will incorporate more comprehensive statistical representations of the fracture geometry, including probabilistic distributions of aperture and size, and expand the experimental design to include a broader range of fracture orientations. These improvements enhanced the realism and applicability of the model to complex geological settings.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Weijie Zhang: Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization. **Yuanlin Bai:** Writing – original draft, Visualization, Validation, Data curation. **Chenghao Han:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis. **Pengfei Wang:** Visualization, Data curation, Conceptualization. **Danyang Zhang:** Supervision, Formal analysis. **Jicheng Zhang:** Writing – original draft, Software.

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